

STA 35C – Homework 6

Submission due: Tue, May 26 at 11:59 PM PT

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Instructions: Upload a **single PDF file** to Gradescope via Canvas (“Homework 6” under “Assignments”). Name it using your UC Davis email ID prefix and homework number, e.g., `dgsong_hw6.pdf`. Include “STA 35C,” your name, and the last four digits of your student ID on the front page. For coding problems, include your R code, relevant output, and requested plots/tables in the PDF. Late submissions receive 0 points; see the syllabus for full submission and late-policy details.

Problem 1: Bootstrap review (20 points total).

You have an integer-valued random variable with an unknown distribution, whose median m you want to estimate. You collect a sample of size 7 from this population, obtaining the following data:

2, 0, 0, 1, 2, 3, 2.

Let $\hat{m}$ be the sample median computed from these 7 observations.

- (a) (6 points) If you resample *with replacement* from this original sample of size 7 to generate a new bootstrap sample of size 7, what is the probability of drawing the same sequence of values (2, 0, 0, 1, 2, 3, 2) in that bootstrap sample?
- (b) (8 points total) Suppose we generated the following five bootstrap samples, each of size 7:

	Bootstrap 1	Bootstrap 2	Bootstrap 3	Bootstrap 4	Bootstrap 5
Sample 1	2	0	2	3	0
Sample 2	2	1	2	0	1
Sample 3	0	0	2	2	0
Sample 4	0	0	3	2	3
Sample 5	2	2	2	2	1
Sample 6	3	2	1	1	2
Sample 7	3	2	3	0	2

- (i) (4 points) Compute the bootstrap estimates of the median for each bootstrap sample.
- (ii) (4 points) Construct a 95% confidence interval for the unknown median m , using:
- $\hat{m}$ estimated from the original sample (2, 0, 0, 1, 2, 3, 2), and
 - the *normal approximation*, with the standard deviation estimated from the 5 bootstrapped median estimates. (*Hint:* $z_{0.975} \approx 1.96$.)
- (c) (6 points) In this context, how should we interpret “95%” in the 95% confidence interval for m ? Specifically, describe succinctly which probability is intended to be approximately 95%.

Problem 2: Regularization & multiple testing review (20 points total).

- (a) (10 points) Recall that lasso regression estimates the coefficients in a linear model by minimizing

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j|.$$

As we increase λ from 0 to ∞ , how do you expect each of the following to behave?

Pick among these five options: (1) "Remain constant," (2) "Steadily increase," (3) "Steadily decrease," (4) "Decrease initially, and then eventually start increasing in a U shape," or (5) "Increase initially, and then eventually start decreasing in an inverted U shape."

Each question is worth 2 points. Briefly justify your choice in one sentence.

- (i) Training error, measured by RSS or MSE
 - (ii) Test error, measured by MSE
 - (iii) (Squared) bias
 - (iv) Variance
 - (v) Irreducible error
- (b) (10 points total) Consider a single null hypothesis H_0 ; the table on the *left* below shows the probabilities of each outcome ($p_1 + p_2 + p_3 + p_4 = 1$). Now suppose we have m (e.g. 100) hypotheses tested simultaneously; let N_1, N_2, N_3, N_4 count each outcome, so $N_1 + N_2 + N_3 + N_4 = m$.

Single	H_0 is true	H_0 is not true	Multiple	H_0 is true	H_0 is not true
Reject H_0	p_1	p_2	Reject H_0	N_1	N_2
Not reject H_0	p_3	p_4	Not reject H_0	N_3	N_4

- (i) (4 points total)
 - (2 points) For a single randomly selected hypothesis, what is the probability that H_0 is rejected (whether true or not)?
 - (2 points) Among m null hypotheses, how many are not true (i.e. should ideally be rejected), and how many are actually rejected?
- (ii) (2 points) A level- α test controls the Type I error probability. Write this requirement as an inequality involving p_1, p_2, p_3, p_4 and α .
- (iii) (2 points) Suppose we aim to control the *familywise error rate* (*FWER*) at level β (e.g. 0.05). Express this goal as an inequality involving N_1, N_2, N_3, N_4 and β .
- (iv) (2 points) Suppose instead we aim to control the *false discovery rate* (*FDR*) at level q (e.g. 0.10). Express this goal as an inequality involving N_1, N_2, N_3, N_4 and q .

Problem 3: Regression & smoothing splines (45 points total + 10 bonus points).

- (a) (15 points) [JWHT21, Chapter 7, Exercise 1].
- (b) (30 points) [JWHT21, Chapter 7, Exercise 9].
(Note: The **Boston** dataset is contained in the **ISLR2** library.)
- (c*) (*10 bonus points) [JWHT21, Chapter 7, Exercise 2].

Problem 4: Principal component analysis (15 points total).

We will explore the basics of PCA conceptually, numerically, and through a brief coding demonstration in **R**, all without heavy linear algebra.

- (a) **(5 points)**. In your own words, explain what principal components are and why PCA can be useful for dimension reduction. Briefly describe how PCA finds directions of maximum variance, verbally or graphically.
- (b) **(5 points)**. Consider the eight points:

$$(1, 2), (2, 1.5), (3, 2.5), (4, 3), (2, 3.5), (3, 4.5), (4, 5), (5, 6).$$

- (i) Compute the coordinate-wise means and subtract them from each point to center the data.
 - (ii) By drawing a scatter plot of these centered points, identify a plausible “best line” of maximum spread. Explain briefly why it seems correct.
 - (iii) Describe how to reduce these 2D points to 1D using only the first principal component.
- (c) **(5 points)**. Using **R**, do the following:
- (i) Create an 8×2 matrix **X** from the eight original data points.
(e.g. `X <- rbind(c(1,2), c(2,1.5), ..., c(5,6))`).
 - (ii) Run PCA using `prcomp(X, center=TRUE, scale=FALSE)` (ensuring you do not scale).
 - (iii) Report the first principal component vector (from `pca_res$rotation`, up to sign) and the percentage of variance explained by it.
 - (iv) In a sentence or two, compare these PCA results to your “best line” direction from part (b).

References

- [JWHT21] Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani. *An Introduction to Statistical Learning: with Applications in R*, volume 112 of *Springer Texts in Statistics*. Springer, New York, NY, 2nd edition, 2021.